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A finite element method to model progressive fracturing

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Abstract

In this paper, a theoretical basis is established for a finite element iterative algorithm for load-displacement analysis of elasticity problems in which the elastic body fractures under load. Location of fractures is determined by testing a Coulomb friction criterion along inter-element edges. Convergence of the algorithm to a physically-meaningful solution is proved, and is illustrated by results for a cantilever-type problem.

1. Introduction

The finite element method (FEM) has proved a powerful and versatile tool for modelling the deformation under loading of structures and continua. Material models used have extended from linear elasticity to cover non-linear elasticity, elasto-plasticity, etc., and from static analyses to quasi-static and dynamic. The FEM can model irregular and complex geometries and loadings. The main restriction of the method, when applied in continuum mechanics, is that the deformed body must remain a continuum without developing separations or discontinuities between elements. Since in reality, loaded bodies do develop cracks and fractures, perhaps leading to total separation into distinct blocks, various methods have been proposed to adapt the FEM. The earliest of these is the joint or interface element of Goodman [1], a zero-thickness element inserted between two finite elements, in which shear stress along the joint may cause relative tangential displacements between corresponding nodes on the two joint faces. The concept has been extended also to allow relative normal displacements, i.e. joint opening and closing, caused by normal stress [2]. As an alternative to the zero-thickness element, Zienkiewicz et al. [3] proposed a thin-layer element, since applied in geomechanics by Desai et al. [4]. Mention should also be made of adaptive mesh refinement to model shear loads, and Moving Finite Elements [5], which work by making the mesh denser along a discontinuity or shock.

Such techniques are however limited in the degree of relative displacement and rotation that can be modelled, since the mesh essentially remains a fully-connected continuum. The problem may be approached from the other direction, so to speak, by starting with a mesh of discrete blocks which can move independently and interact through contact – this is the basis of the discrete element method (DEM) (see e.g. [6]). These discrete element blocks may themselves be rigid [7], or deformable: e.g. Discontinuous Deformation Analysis [8] or Discrete Finite Element Method [9]. Under this heading, the Rigid Bodies – Spring Models (RSBM) approach may also be classed [14, 15], in which rigid blocks are connected by elasto-plastic springs distributed over their common surface; these springs can also become cut, causing complete separation. This method has produced reasonable results for single-crack

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problems, while the DEM is suitable for modelling objects such as ice sheets which rapidly develop multiple fractures throughout the whole body.

A drawback, common to both the joint-element and discrete block approaches, is that the locations of the discontinuities must be fixed in advance, either by the insertion of joint elements in the mesh at predetermined locations, or by initial discretization of the mesh into discrete blocks. The subsequent deformation under loading is therefore dependent upon the pattern of discontinuities introduced. Admittedly, the joint elements can be made to represent potential discontinuities, with relative displacements occurring only if Coulomb friction or normal tension criteria are satisfied [10]. However, it would be computationally expensive to install such elements at every interface between finite elements in a refined mesh, to allow for all possible patterns of progressive fracturing of an initially continuous body. There are important practical situations in which such progressive fracturing occurs, for example in the rock layer forming the unsupported roof of a deep-level underground excavation – so-called ‘blocky’ rock behaviour. Here, the part of the rock stratum not overlying the excavation remains continuous, while within the part forming the excavation roof there will be a region where the rock fractures propagate, leading to separation into discrete blocks. The boundary between the continuous and the fractured regions of the rock stratum is not known in advance, although there are analytical techniques for prediction.

In this paper, the mathematical basis for a finite element method to model this type of situation will be established. The algorithm starts from an initially continuous mesh (perhaps containing some initial separations between elements), and proceeds iteratively. At each iteration the stresses on all inter-element edges are tested against a Mohr–Coulomb fracture criterion. Where this criterion is satisfied, separations are introduced, and the problem re-solved. In the next section, an algorithm for determining a solution involving created fractures is described. The idea of this algorithm is similar to that of the static relaxation algorithm sometimes used in the DEM [11]; a dynamic algorithm is also possible, but such practical aspects will be dealt with in a separate publication. Section 3 establishes the mathematical model, and in Section 4 it is shown that for a properly-posed problem, the algorithm of Section 2 will lead to a physically meaningful solution. A numerical example is given in Section 5.

2. Formulation of the method

For simplicity, we consider a single elastic body which deforms and fractures under load. Let Ω be the reference configuration of the elastic body. Let Γ be the original boundary of $\Omega \subset \mathbb{R}^n$, ($n = 2, 3$), i.e. the boundary which corresponds to the surface of the elastic body before any fractures occur. Let Γ_0 be the fixed boundary. Let f be the body force and g be the traction which is applied on Γ_1 ($\Gamma_0 \cup \Gamma_1 = \Gamma$).

As in the FEM, we introduce a mesh $\mathcal{T}$ in Ω . By applying the usual FEM to the boundary value problem of static elasticity, we obtain a FEM displacement solution u_0 which is continuous in Ω . New boundaries of Ω , i.e. fractures in the elastic body, can now be introduced. For simplicity, we assume that fractures are only introduced along $(n - 1)$ -dimensional mesh faces where the fracture criterion of the material is satisfied. We denote the resulting boundary of Ω as $\partial\Omega_0$. In other words, $\partial\Omega_0 \setminus \Gamma$ corresponds to the fractures produced by the solution u_0 . We may divide the domain Ω into two parts, Ω_{con} and Ω_{dis} , by defining

$$\Omega_{\text{con}} = \{\text{all elements } K \text{ satisfying } \text{dist}(K, \partial\Omega_0 \setminus \Gamma) > r\}, \quad (2.1)$$

$$\Omega_{\text{dis}} = \{\text{all elements } K \text{ satisfying } \text{dist}(K, \partial\Omega_0 \setminus \Gamma) \leq r\}, \quad (2.2)$$

where $r > 0$ is a constant and $\text{dist}(\cdot, \cdot)$ represents the distance between the two sets. We call each connected domain of Ω_{con} and each connected part of Ω_{dis} a block, denoted by $B_1, B_2, \dots, B_N$.

Without loss of generality, we may assume that $\partial\Omega_0 \setminus \Gamma \neq \emptyset$. Thus, the displacement field u_0 is no longer an equilibrium state of the problem with the new boundary $\partial\Omega_0$. The following iterative scheme may be applied to carry on the solution procedure, i.e. to find a displacement field and boundary that will be in equilibrium. At the $(IN + k)$ th step, all nodes except those belonging to the block B_k are

fixed, a solution Δu_{IN+k} for the reduced boundary value problem of static elasticity (see below) is found, and the global displacement field is updated by adding Δu_{IN+k} to its previous value, i.e.

$$u_{IN+k} = u_{IN+k-1} + \Delta u_{IN+k}, \quad (2.3)$$

while the boundary of Ω is also updated by adding new boundaries corresponding to new fractures produced by the incremental displacement Δu_{IN+k} , and the result is denoted by $\partial\Omega_{IN+k}$.

The relaxation process is continued until either

- (1) a satisfactory equilibrium state is obtained; or
- (2) $\text{dist}(\Omega_{\text{con}}, \partial\Omega_n \setminus \Gamma) = 0$ (i.e. a fracture has arisen in the continuum part of Ω); or
- (3) failure occurs, i.e. the reduced boundary value problem is not properly posed.

In case (1), u_n is taken as the approximate solution to the problem. In case (2), Ω_{con} and Ω_{dis} are redefined by

$$\Omega_{\text{con}} = \{\text{all elements } K \text{ satisfying } \text{dist}(K, \partial\Omega_n \setminus \Gamma) > r\}, \quad (2.4)$$

$$\Omega_{\text{dis}} = \{\text{all elements } K \text{ satisfying } \text{dist}(K, \partial\Omega_n \setminus \Gamma) \leq r\}, \quad (2.5)$$

and the iterative relaxation process can be continued again. Case (3) happens when a block B_k becomes a distinct body and has no contact force (or too few contact points) at its edges or corners [12] to keep it at an equilibrium state. To let the calculation continue, we may let the element move and deform along the steepest descent direction for the potential energy of the reduced system. The displacement is such that the potential energy is actually lowered and the 'time' for the element to have such displacement should be in a reasonable interval. This 'quasi-static' algorithm will not be pursued further here.

When the loads f and g are large, an incremental method can be used, i.e. the loads are imposed incrementally and in each incremental step the iterative scheme described above is carried out.

We now consider the reduced boundary value problem of static elasticity for the block B_k . There are several situations:

- (a) B_k is a connected domain of Ω_{con} ;
- (b) B_k is a connected domain of Ω_{dis} , but $\text{dist}(B_k, \partial\Omega_{k-1} \setminus \Gamma) > 0$;
- (c) $\text{dist}(B_k, \partial\Omega_{k-1} \setminus \Gamma) = 0$, but B_k is not a distinct body;
- (d) B_k is a distinct body.

In situations (a) and (b), the reduced problem is simply the finite element boundary value problem; we only discuss the situations (c) and (d). To reduce the complexity, it is necessary to simplify the contact model of blocks. In the simplest model (see [11, 12]), overlaps are admitted on contact interfaces and the interaction forces between blocks are assumed to be linearly dependent on the depth of the overlaps and the distance of slide. In the present model, interaction forces between two blocks are the combination of two different kinds of force; one is the force produced by the stiffness of blocks which have contact with the block in question, and the other is the force produced by overlapping and sliding. The forces of the latter kind, which are introduced artificially to force convergence to a realistic solution, will eventually tend to zero. Thus, for the block B_k in situations (c) and (d), the reduced problem has the form

$$A_k(u_{IN+k-1}) \Delta u_{IN+k} = \sum f_{kj}^1 + \sum f_{kj}^2 + f_k^3 - F_k(u_{IN+k-1}) \quad (2.6)$$

where f_{kj}^1 depend on the stiffness of the block B_j and the relative position of B_j and B_k , f_{kj}^2 depend only on the overlap and relative movement of B_k and B_j , f_k^3 is the load imposed on B_k and $F_k(u_{IN+k-1})$ is the equivalent balance load for the displacement field u_{IN+k-1} . $A_k(u_{IN+k-1})$ is the submatrix of the global stiffness matrix, obtained by selecting the entries corresponding to the degrees of freedom of the nodes belonging to B_k , subject to displacements u_{IN+k-1} .

3. Mathematical model

We consider the static problem of an elastic body under load. Let $\Omega \subset R^n$ ($n = 2, 3$) be the reference configuration of the elastic body. Let $W(x, F)$ be the stored energy function, where $F = I + \nabla u$. Let f be

the density of the body force, and g be the density of traction acting upon the elastic body through the boundary Γ_1 . Assume that a coercive boundary condition is given on Γ_0 by

$$u(x) = u_0(x), \quad \text{for } x \in \Gamma_0.$$

Then, the total potential energy of the elastic body can be written as

$$E(u) = \int_{\Omega} W(x, F) \, dx - \int_{\Omega} f \cdot u \, dx - \int_{\Gamma_1} g \cdot u \, dx. \quad (3.1)$$

In continuum mechanics, the equilibrium state of the problem is given by the minimum of the functional $E(u)$ in a properly defined admissible function space, such as

$$\mathcal{A} = \{u \in W^{1,p}(\Omega) \mid \det F > 0, \text{ a.e. in } \Omega; u = u_0, \text{ on } \Gamma_0\} \quad (3.2)$$

where $\det F$ is the determinant of F and a.e. denotes 'almost everywhere'. But in fracture mechanics, the body may produce some fractures under load according to certain criteria. Here, we assume that the Mohr–Coulomb criterion [13] applies, i.e. fracture will occur at a point in Ω with stress σ if

$$\sigma_1(1 - \sin \phi) - \sigma_3(1 + \sin \phi) - 2c \cos \phi = 0, \quad (3.3)$$

where σ_1 is the major principal stress, σ_3 is the minor principal stress and ϕ, c are material constants (angle of friction and cohesion). The fracture occurs on the failure plane of which the normal makes an angle of $\pi/4$ with the direction of σ_1 and σ_3 . To reflect the feature, an admissible function should satisfy the restriction

$$\sigma_1(1 - \sin \phi_0) - \sigma_3(1 + \sin \phi_0) - 2C_0 \cos \phi_0 < 0, \quad \text{everywhere in } \Omega \quad (3.4)$$

Since the pointwise values of σ_1 and σ_3 are meaningless in $\mathcal{A}$, the values in (3.4) are defined by

$$\sigma_i(x) = \lim_{r \rightarrow 0} \frac{1}{S(r)} \int_{|y-x| \leq r} \sigma_i(y) \, dy, \quad i = 1, 3,$$

with

$$S(r) = \int_{|x| \leq r} dx$$

It may happen that there is no minimum of $E(u)$ in $\mathcal{A}$ subject to (3.4). If this is the case, it means that a solution must involve some discontinuities in Ω , i.e. there must have been some fractures produced in the body when it reaches the equilibrium state. Since we have no idea where these discontinuities will be located, we have to consider all possibilities.

Let $\{\Omega_\alpha\}$ be a set of domains satisfying

$$\begin{aligned} \Omega_\alpha &\subset \Omega, \\ \partial\Omega_\alpha &\supset \partial\Omega, \\ \Omega_\alpha \cup (\partial\Omega_\alpha \setminus \partial\Omega) &= \Omega. \end{aligned}$$

Define

$$\mathcal{A}_\alpha = \{u \in W^{1,p}(\Omega_\alpha) \mid \det F > 0, \text{ a.e. in } \Omega_\alpha; u = u_0 \text{ on } \Gamma_0\}, \quad (3.5)$$

$$\mathcal{A}_\alpha^c = \{u \in \mathcal{A}_\alpha \mid u \text{ satisfies (3.4) in } \Omega_\alpha\}, \quad (3.6)$$

$$\tilde{\mathcal{A}}_\alpha^c = \{u \in \mathcal{A}_\alpha \mid u \text{ satisfies (3.3) or (3.4) in } \Omega_\alpha\}, \quad (3.7)$$

$$E_\alpha(u) = \int_{\Omega_\alpha} W(x, F) \, dx - \int_{\Omega_\alpha} f \cdot u \, dx - \int_{\Gamma_1} g \cdot u \, dx. \quad (3.8)$$

DEFINITION 3.1. Suppose $u_\alpha \in \mathcal{A}_\alpha^c$ is a local minimum of E_α in $\mathcal{A}_\alpha^c$. Then u_α is defined to be a solution of the static equilibrium problem of elasticity subject to the Mohr–Coulomb criterion (3.4).

DEFINITION 3.2. Let

$$\hat{\mathcal{A}}^c = \bigcup_{\alpha} \tilde{\mathcal{A}}_{\alpha}^c,$$

Then $u_{\alpha} \in \tilde{\mathcal{A}}_{\alpha}^c$ and $u_{\beta} \in \tilde{\mathcal{A}}_{\beta}^c$, $\Omega_{\alpha} \subset \Omega_{\beta}$ are defined to be equivalent in $\hat{\mathcal{A}}^c$ if

$$u_{\beta} \in \tilde{\mathcal{A}}_{\beta}^c \setminus \tilde{\mathcal{A}}_{\beta}^c \quad \text{and} \quad u_{\beta}|_{\Omega_{\alpha}} = u_{\alpha}.$$

DEFINITION 3.3. Define a functional $\hat{E}: \hat{\mathcal{A}}^c \rightarrow R^1$ by

$$\hat{E}(u) = E_{\alpha}(u), \quad \text{if } u \in \tilde{\mathcal{A}}_{\alpha}^c.$$

LEMMA 3.1. For any $u \in \hat{\mathcal{A}}^c$, there is a Ω_{β} such that $u \in \tilde{\mathcal{A}}_{\beta}^c$.

PROOF. By Definition 3.2, there is a Ω_{α} such that $u \in \tilde{\mathcal{A}}_{\alpha}^c$. Let

$$\partial u^c = \{x \in \Omega_{\alpha} \mid u \text{ does not satisfy (3.4) at } x\}.$$

Let $\Omega_{\beta} \in \{\Omega_{\alpha}\}$ be such that

$$\partial \Omega_{\beta} = \partial \Omega_{\alpha} \cup \partial u^c.$$

Then $u|_{\Omega_{\beta}} \in \tilde{\mathcal{A}}_{\beta}^c$. QED.

THEOREM 3.1. $u \in \hat{\mathcal{A}}^c$ is a solution to the static equilibrium problem of elasticity subject to Mohr–Coulomb criterion (3.4) if and only if u is a local minimum of $\hat{E}$ in $\hat{\mathcal{A}}^c$.

PROOF. The conclusion follows directly from Definition 3.1 and Lemma 3.1. QED.

Generally speaking, as in many non-linear problems, there is no unique solution. To determine the true final equilibrium state, we need to know the deformation history, which can often be determined by the loading history and the original state, as well as the material properties of the body.

4. Convergence of the method

Let $\mathcal{T}$ be a mesh in $\Omega \subset R^n$, $n = 2, 3$. Suppose fractures can only occur on $(n-1)$ -dimensional mesh faces. Then we have a subset of $\{\Omega_{\alpha}\}$ which contains only a finite number of domains $\{\Omega_k\}_{k=0}^{K(\mathcal{T})}$, where $\Omega_0 = \Omega$ and $K(\mathcal{T})$ is a constant depending on $\mathcal{T}$. Let $\mathcal{A}_k(\mathcal{T})$, $k = 0, 1, \dots, K(\mathcal{T})$ be finite element function spaces properly defined on Ω_k with the mesh $\mathcal{T}$ such that $\mathcal{A}_k(\mathcal{T}) \subset \mathcal{A}_k$. Define

$$\mathcal{A}_k^c(\mathcal{T}) = \{u \in \mathcal{A}_k(\mathcal{T}) \mid u \text{ satisfies (3.4) on } \mathcal{T} \text{ in } \Omega_k\},$$

$$\tilde{\mathcal{A}}_k^c(\mathcal{T}) = \{u \in \mathcal{A}_k(\mathcal{T}) \mid u \text{ satisfies (3.4) or (3.3) on } \mathcal{T} \text{ in } \Omega_k\},$$

where ‘on $\mathcal{T}$ in Ω_k ’ means ‘on all $(n-1)$ -dimensional faces of $\mathcal{T}$ in Ω_k ’. Define

$$\hat{\mathcal{A}}^c(\mathcal{T}) = \bigcup_{k=0}^{K(\mathcal{T})} \tilde{\mathcal{A}}_k^c(\mathcal{T})$$

with the topology induced from $\hat{\mathcal{A}}^c$.

DEFINITION 4.1. Suppose $u \in \hat{\mathcal{A}}^c(\mathcal{T})$ is a local minimum of $\hat{E}$ in $\hat{\mathcal{A}}^c(\mathcal{T})$. Then u is called an approximate solution of the static equilibrium problem of elasticity subject to the Mohr–Coulomb criterion (3.4).

By a similar argument to that in Section 3, we have

THEOREM 4.1. $u \in \hat{\mathcal{A}}^c(\mathcal{T})$ is an approximate solution of the static equilibrium problem of elasticity subject to the Mohr–Coulomb criterion (3.4) if and only if there exists a k such that u is a local minimum of E_k in $\mathcal{A}_k^c$.

THEOREM 4.2. Assume the stored energy function $W(x, F)$ is continuous and for $F_l, l = 1, 2, \dots$, with $\det F_l > 0$, satisfies

$$\lim_{l \rightarrow \infty} W(x, F_l) \rightarrow +\infty, \begin{cases} \text{if } \lim_{l \rightarrow \infty} \det F_l = 0, \\ \text{or } \lim_{l \rightarrow \infty} \|F_l\| \rightarrow +\infty. \end{cases} \quad (4.1)$$

Assume the displacement field is such that

$$\|u\| \leq C_1 + C_2 \|F\| \quad (4.2)$$

for some constants $C_1, C_2 > 0$. Then $\hat{E}$ has at least one local minimum in $\hat{\mathcal{A}}^c(\mathcal{T})$.

PROOF. Let Ω_D be such that $\partial\Omega_D \setminus \partial\Omega_0$ contains all $(n-1)$ -dimensional mesh faces of $\mathcal{T}$. Then we have

$$\mathcal{A}_D^c(\mathcal{T}) = \mathcal{A}_D(\mathcal{T}). \quad (4.3)$$

Let $\{u_l\} \subset \mathcal{A}_D(\mathcal{T})$ be a minimizing sequence of $\hat{E}$ in $\mathcal{A}_D(\mathcal{T})$. By (4.1) and (4.2), $\{\|u_l\|\}$ and $\{\|F_l\|\}$ are uniformly bounded. Since $\mathcal{A}_D(\mathcal{T})$ is of finite dimension, there exists a subsequence of $\{u_l\}$, still denoted by $\{u_l\}$, and a $u_D \in \mathcal{A}_D(\mathcal{T})$ such that

$$\begin{aligned} u_l &\rightarrow u_D, \quad \text{as } l \rightarrow \infty; \\ F_l &\rightarrow F_D, \quad \text{as } l \rightarrow \infty. \end{aligned}$$

Thus, it follows from the continuity of $W(\cdot, \cdot)$ that

$$\hat{E}(u_D) = \lim_{l \rightarrow \infty} \hat{E}(u_l) = \inf_{u \in \mathcal{A}_D(\mathcal{T})} \hat{E}(u) < +\infty. \quad (4.4)$$

By (4.4) and (4.1), we have

$$\lim_{l \rightarrow \infty} \det F_l = \det F_D > 0.$$

Hence

$$u_D \in \mathcal{A}_D(\mathcal{T}). \quad (4.5)$$

From (4.3) and (4.4) and (4.5) it follows that $u_D \in \mathcal{A}_D^c(\mathcal{T})$ is a minimum of $\hat{E}$ in $\mathcal{A}_D^c(\mathcal{T})$. The theorem follows now from Definition 4.1 and Theorem 4.1. QED.

In most cases, there are many local minima of $\hat{E}$ in $\hat{\mathcal{A}}^c(\mathcal{T})$, and the solution u_D given in the proof of Theorem 4.2, which can be considered as a solution obtained by the DEM, is not a realistic numerical solution of the original problem, in which the body is initially a continuum. However, the algorithm described in Section 2 provides an approach by which a more realistic solution can be obtained. The following theorem proves that this algorithm will, under certain assumptions, lead to a local minimum of $\hat{E}$ in $\hat{\mathcal{A}}^c(\mathcal{T})$; this will be a more realistic solution than that from the DEM, since the fracturing has developed progressively rather than being preordained.

THEOREM 4.3. Let $\{u_l\}$ and $\{\Omega_l\}$ be sequences defined by the algorithm (see Section 2). With the same assumptions as in Theorem 4.2, there exists a $l_0 \geq 0$ and an $\Omega_l \in \{\Omega_k\}_{k=0}^{K(l)}$ and an $u_h \in \mathcal{A}_l^c(\mathcal{T})$ such that

$$\Omega_l = \Omega_{l_0}, \quad \text{for } l \geq l_0; \quad (4.6)$$

$$u_l \rightarrow u, \quad \text{as } l \rightarrow \infty. \quad (4.7)$$

further, u_h is a local minimum of $\hat{E}$ in $\hat{\mathcal{A}}^c(\mathcal{T})$.

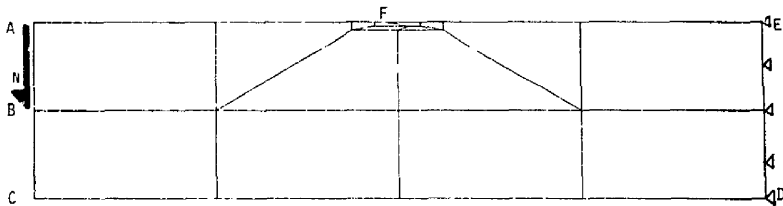


Fig. 1. Cantilever mesh.

PROOF. Since $\{\Omega_l\} \subset \{\Omega_k\}_{k=0}^{K(l)}$ and $\dots \Omega_{l-1} \subset \Omega_l \subset \Omega_{l+1} \dots$, there can only be finite different subscripts l_i such that $\Omega_{l_{i-1}} \neq \Omega_{l_i}$. Let l_0 be the largest such subscript, let $\Omega_l = \Omega_{l_0}$, then (4.6) holds.

For $l \geq l_0$, we have $u_l \in \mathcal{A}_{l_0}^*(\mathcal{T})$. Since $E_{l_0}(u_l)$ are decreasing, by a similar argument to that in Theorem 4.2, we can find a $u_h \in \mathcal{A}_{l_0}^*(\mathcal{T})$ and a subsequence $\{u_l\}$ such that (4.7) holds. QED.

To prove that an accumulation point u of $\{u_h\}$ as $h \rightarrow 0$ is a local minimum of $\hat{E}$ in $\mathcal{A}^c$, we need some additional assumptions. The simplest such assumption is that $W(x, \cdot)$ is convex. We shall not discuss this in this paper. In practice, the algorithm usually leads to a local minimum.

To admit fractures to be introduced within elements will certainly give better approximate solutions, but it makes the program much more complicated. In many applications, the directions of fractures are

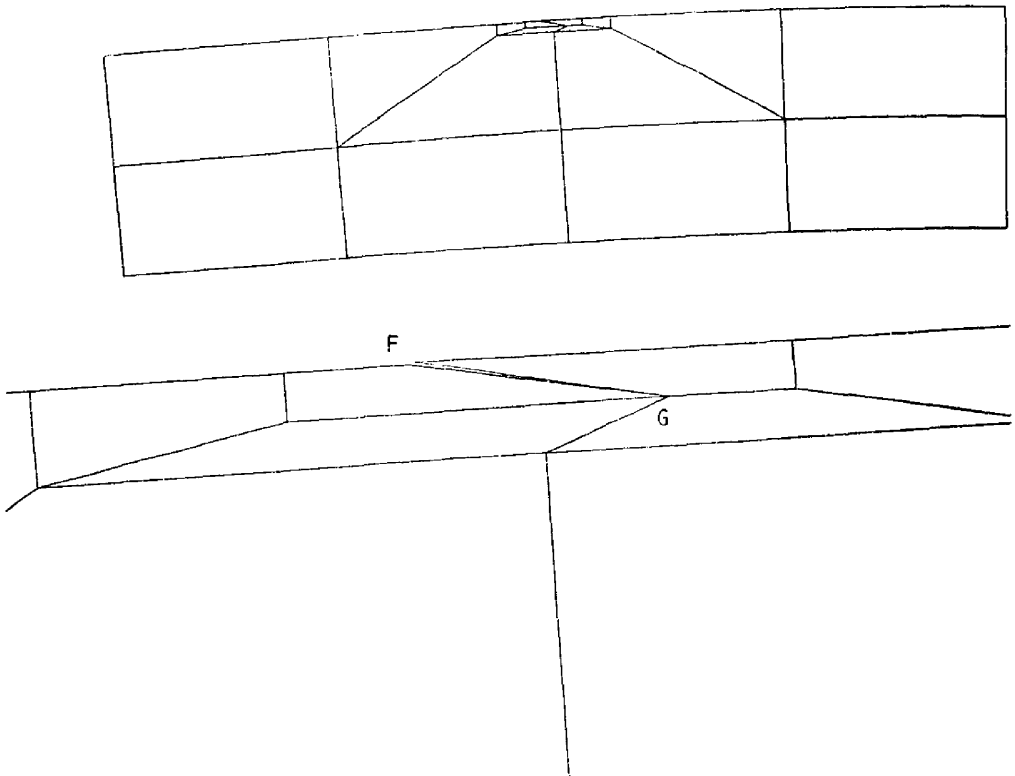


Fig. 2. Mesh after 1st iteration: (a) full mesh, (b) zoom.

known in advance. In such situations, we can always choose the mesh to avoid the need to introduce fractures within elements.

5. Numerical example

The progress of the algorithm in modelling progressive fracturing of an elastic body, is illustrated in the following cantilever-type example. A thick-plate cantilever is modelled in plane strain by the finite element mesh shown in Fig. 1. The elements are eight-noded Serendipity quadrilaterals. The body is fixed, as if to a wall, along the side DE, and is loaded by a load $N = 0.2$ acting vertically downwards along the side AB. The material has elasticity constants $E = 500 \text{ N/m}^2$ and $\nu = 0.3$.

An initial fracture exists along the inter-element edge FG. Fig. 2 illustrates the mesh deformation after one iteration of the algorithm and an opening of the initial fracture along FG is seen.

The Coulomb criterion applied on element edges to determine if fracturing occurred, used a cohesion $c = 2.0$ and angle of friction $\phi = 4.0^\circ$. After the first iteration, this criterion was exceeded along edge GH, and a new fracture developed there, the opening of which can be seen after the second iteration, as shown in Fig. 3.

Subsequent development of the fracturing is shown in Figs. 4 and 5. In Fig. 5(b), the sliding contact and slight interpenetration between two separate blocks of the mesh is apparent.

Interpenetration also occurs at point I in Fig. 6, and this is resolved by a further iteration, producing the mesh in Fig. 7, where element J has become a totally distinct body. For further analysis, a dynamic algorithm would be needed, as mentioned in Section 2.

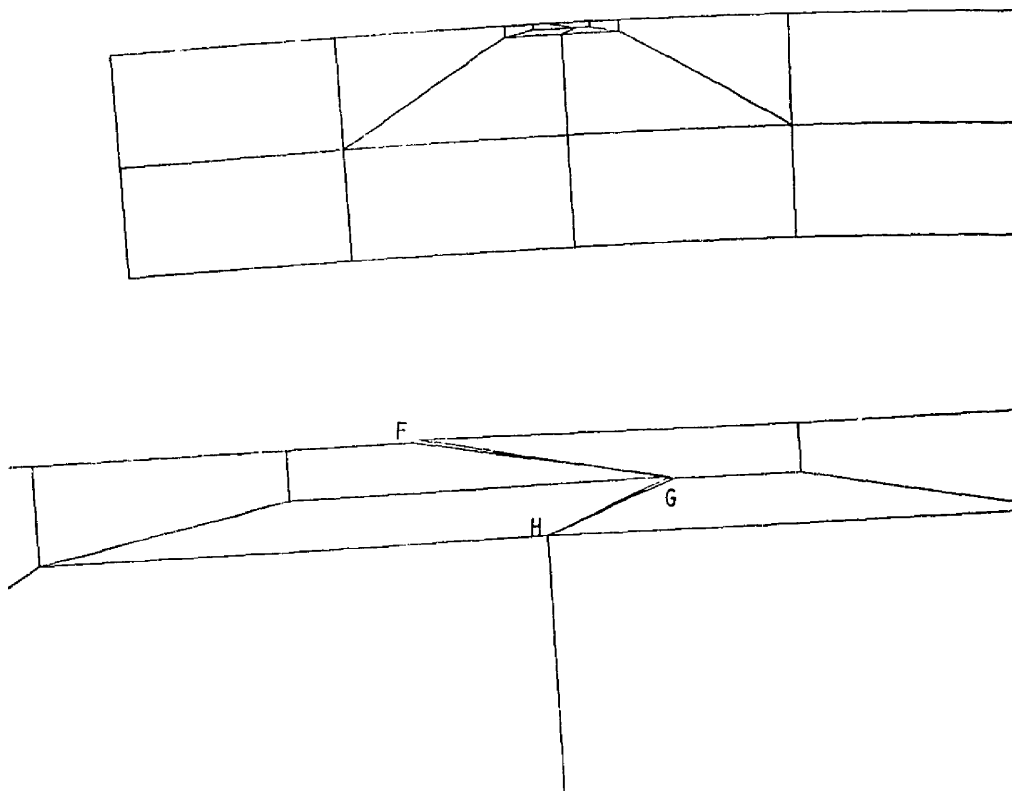


Fig. 3. Mesh after 2nd iteration: (a) full mesh, (b) zoom.

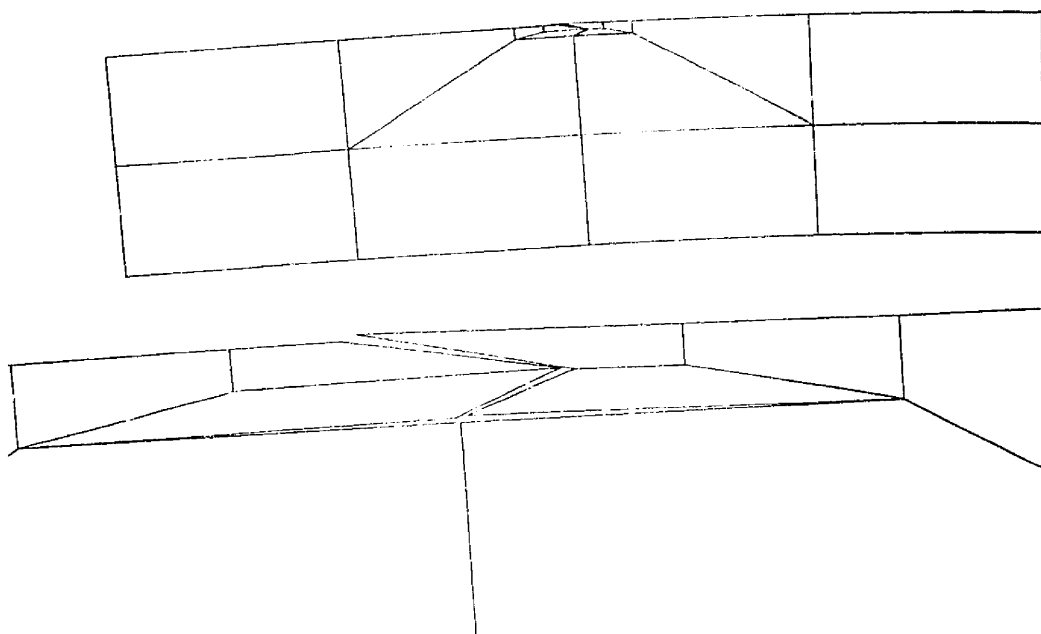


Fig. 4. Mesh after 3rd iteration: (a) full mesh, (b) zoom.

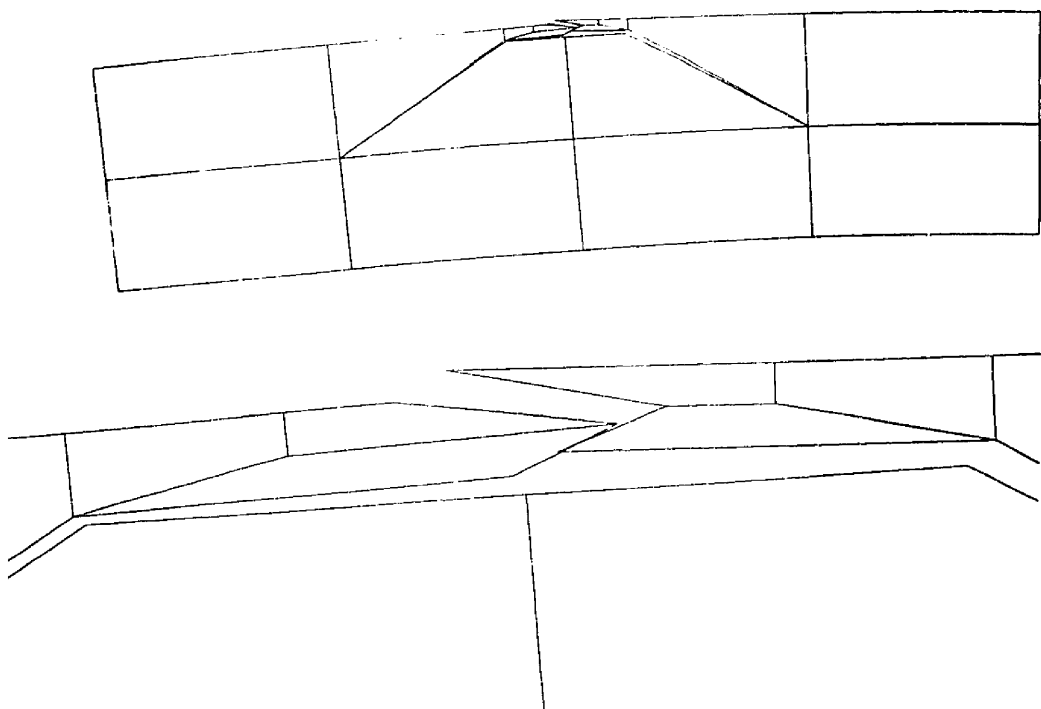


Fig. 5. Mesh after 4th iteration: (a) full mesh, (b) zoom.

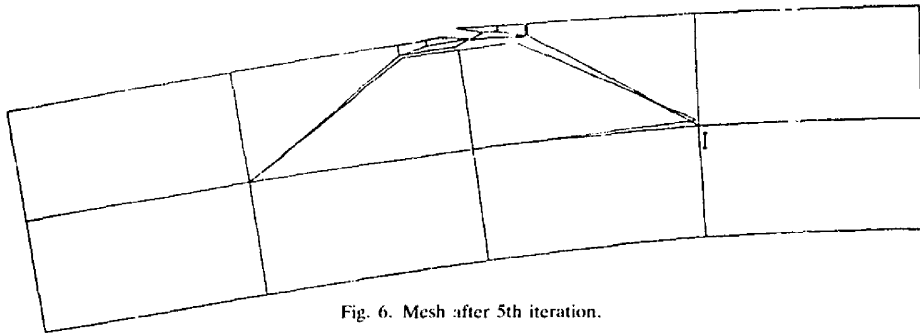


Fig. 6. Mesh after 5th iteration.

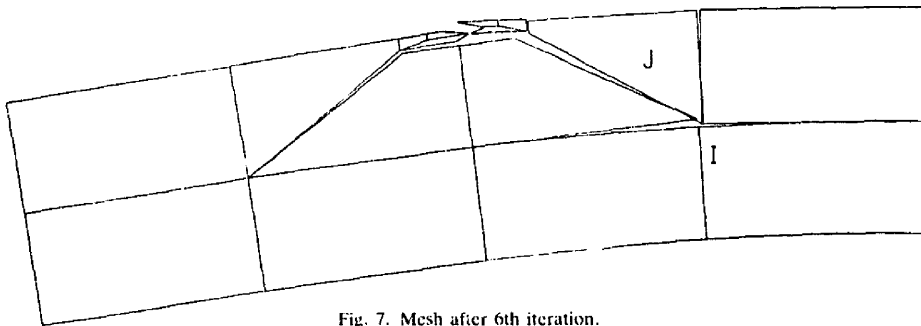


Fig. 7. Mesh after 6th iteration.

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